

Lab 7

- (1) Give the definitions of the following classes: Class P and Class NP.
- (2) Give the definitions of the following sets: PATH, RELPRIME, SAT, HAMPATH, COMPOSITES, CLIQUE. Your definition should be self-contained (e.g. for CLIQUE, explain what a k-clique is).
- (3) Give the definitions of the following classes:
Big-O notation,
Small-O notation,
A verifier,
- (4) Prove that $\text{PATH} \in \text{P}$.
- (5) Prove that CLIQUE is in NP. (give a verifier or a nondeterministic polynomial time Turing machine that decides CLIQUE)